

CS230 DLDCA Mid-Sem, Tue 17 Sep 2024, 13.30-15.30pm, Max. Marks: 30**General instructions**

- Write only in the space provided. Answer briefly but crisply (not lengthily or loosely).
- You are allowed to refer to your own hand-written notes only.
- Write neatly and clearly. Up to +2 **HP** for neat handwriting, neat/crisp answers.
- Answers generally have to be (briefly) explained. State any necessary assumptions.

1. [1 x 2 = 2 marks] Short answer questions:

- a) Mention any two advantages of dynamic linking.

Smaller exe file (better use of memory)

Libraries can be updated for bugfix or performance improvement, without changing existing programs

- b) In a MIPS32 program, a beq instruction is at location 80,000 (decimal value of byte address). In its machine code, the immediate value is -200 (decimal value again). What is the branch target address (which is executed when branch is taken)?

Imm value is taken as word offset from PC+4

So branch target is $80000 + 4 + (4 \times -200) = 79204$ (decimal value)

2. [2 marks] Prove that $f(x,y,z) = xy + yz + zx$ is self-dual using algebraic manipulations (i.e., Boolean Algebra rules and theorems) only. Show all your steps.

See at the end

3. [1 x 4 = 4 marks] MIPS32 program P.exe is generated from assembly files p1.s & p2.s, it has no other external library. MIPS32 program Q.exe is generated from 2 assembly files q1.s and q2.s as well as a statically linked external library lib1.o. There is no dynamically linked library. Answer the following.

- a) In P.exe, suppose registers \$at (\$1) and \$t0 (\$8) are exchanged in all instructions in which they appear, in the machine code. Would P.exe continue to work? Justify.

MIPS32 machine hardware has no special meaning for \$at vs \$t0. So P.exe will continue to work. (This has nothing to do with caller-callee saving conventions).

- b) Answer the same as above with justification, for Q.exe.

Same answer and reason as for P.exe

- c) In P.exe, suppose registers \$s0 (\$16) and \$ra (\$31) are exchanged in all instructions in which they appear, in the machine code. Would P.exe continue to work? Justify.

MIPS32 machine hardware does have a special meaning for \$ra. The jal instruction saves the return address PC+4 onto \$ra. So swapping will NOT work. (This also has nothing to do with caller-callee saving conventions).

- d) Answer the same as above with justification, for Q.exe.

Same answer and reason as for P.exe

4. [2 marks] **Design decision using computer performance:** Suppose that the MIPS32 designers are considering the inclusion of an instruction called add3, which adds 3 registers instead of two added by the current add instruction. Using simulation on a benchmark program, they found that when add3 is used, 5% of the executed instructions use add3. The designers also determined that, with inclusion of add3, the clock cycle length of a single cycle implementation increases from 500ps to 550ps. (1 pico-second = 10^{-12} seconds). Is the inclusion of add3 beneficial?

With add3, execution time $T1 = N \times 1 \times 550$ ps (CPI=1, N=executed instructions)

Without add3, each add3 will be replaced with two add instructions, so number of executed instructions will be $N \times 1.05$

So, without add3, execution time $T2 = N \times 1.05 \times 1 \times 500$

Clearly, $T1$ is larger than $T2$, so inclusion of add3 is NOT beneficial

Name: _____ Roll number: _____

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- ✓ 5. **[3+2=5 marks]** A new flip-flop, called MN flip-flop, is constructed from a JK flip-flop as follows:

$$J = M; K = N \oplus M$$

- a) Derive the characteristic table and excitation table for this new flip-flop

[See at the end](#)

- b) Construct a D flip-flop from this new flip-flop

[See at the end](#)

✓ 6. **[1+2+2=5 marks]** Construct a counter for the following sequence using T flip-flops:

000 → 100 → 111 → 010 → 011

a) Draw the state-table.

[See at the end](#)

b) Construct the present-state next-state table with T-flip-flop as the memory element.

[See at the end](#)

c) Derive the inputs of the T flip-flops.

[See at the end](#)

7. [2+4+2+2=10 marks] Consider the single cycle implementation for the ISA subset {add, sub, and, or, lw, sw, beq} (as in the videos). We are now going to add support for a new instruction (this is not really a MIPS instruction though) called `sw0_inc`. This instruction is the same as `sw` except for two differences: it always uses offset as 0, and it has the side effect of incrementing the value of the base register by a given 16-bit immediate operand.

a) What instruction format can `sw0_inc` use? Draw it, along with brief justification.

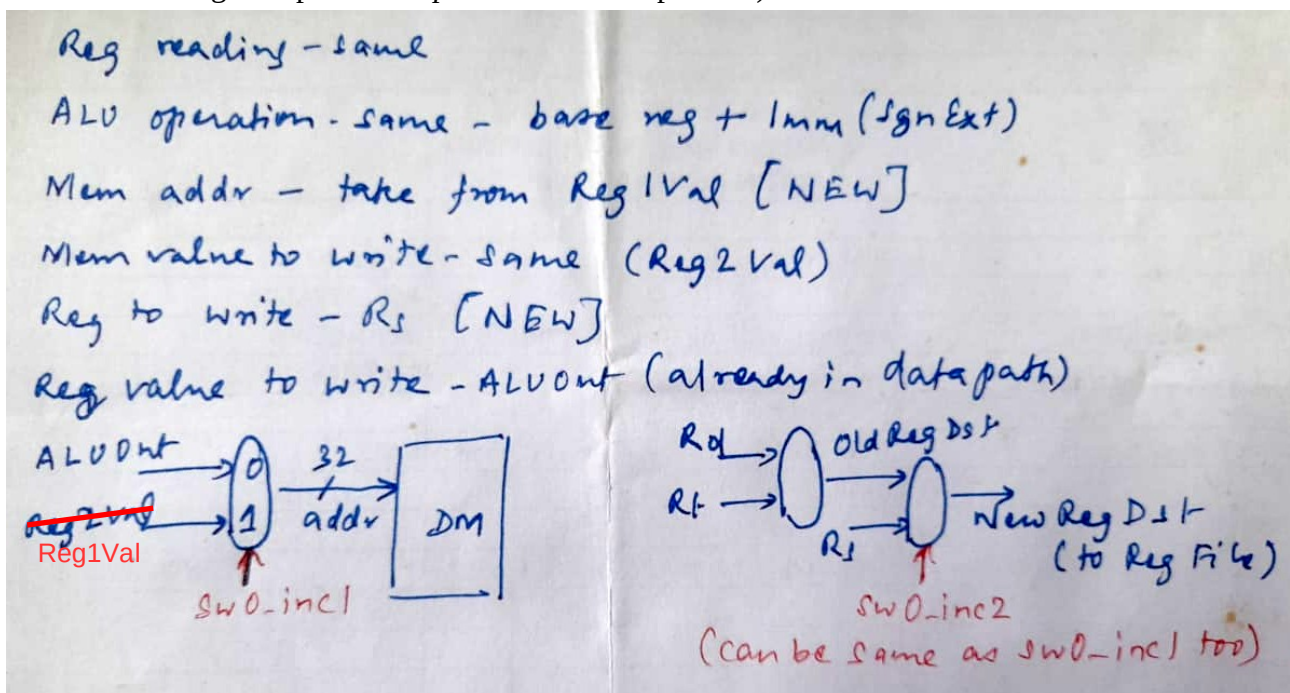
It can use exactly the same format as `sw` (I format)

R_s is base register, R_t is register to be saved to memory

Note that now R_s will have to be written back while executing the instruction

Opcode 6	R_s 5	R_t 5	Immediate 16
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- b) Draw *only* the datapath changes from the original datapath for the ISA subset. Your changes should be *minimal*, adding only muxes as necessary: *avoid* additional ALU/adder or changes to the register file or memory components (it is of course ok to change the possible *inputs* to these components).



- c) Draw *only* the control signal changes from the original ISA subset, i.e. show additional row(s)/column(s) in the truth table.

New row in truth table for sw0_inc instruction:

RegWr=1, RegDst=x, Mem2Reg=0 (choose ALUOut)

sw0_inc1 = sw0_inc2 = 1

ALUSrc=1 (SgnExtImm16), ALUOp = add

MemRd=0, MemWr=1, Branch=0

New cols in truth table for sw0_inc1/sw0_inc1

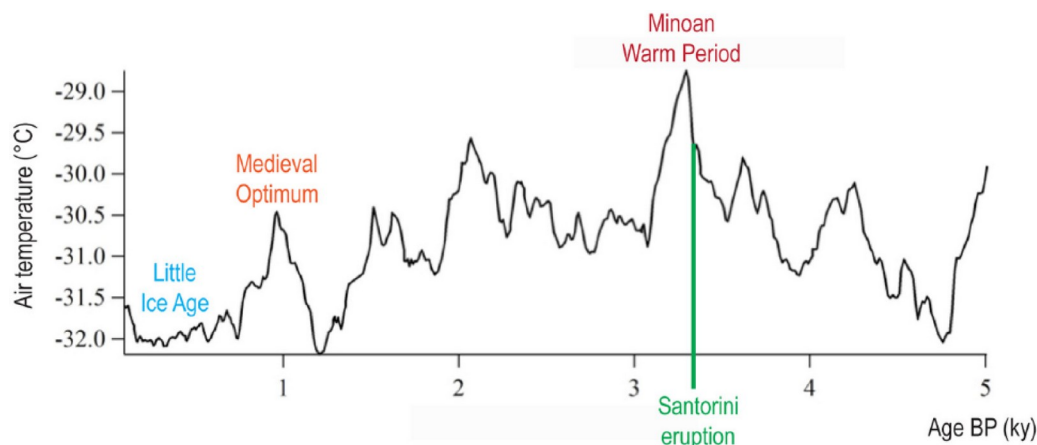
sw0_inc1=0 for lw and sw, sw0_inc1=1 for sw0_inc instruction, sw0_inc1=x for other instructions

sw0_inc2=0 for all reg-reg instructions, and for lw; sw0_inc2=1 for sw0_inc instruction, sw0_inc2=x for sw, beq

- d) Can a lw0_inc instruction be similarly implemented without any change to the main components? Why or why not?

No. Implementing lw0_inc will require two write ports for the register file component.

8. **[Optional, up to 10HP]** The following graph is the estimated Greenland temperature, from the GISP2 (Greenland Ice Sheet Project 2) data, from a 2018 publication. The x-axis is in units of kilo-years Before Present (BP), where BP=0 is taken as 1950AD. As additional data (not in graph), the global average warming since 1950 is about 0.75 °C.



- How much warmer/colder was Greenland during the Medieval Warm Period (1000AD) compared to today? $-(1-0.75) = 0.25$ °C warmer than today approximately _____
- How much warmer/colder was Greenland during the Roman Warm Period (0 AD) compared to today? $-(2-0.75) = 1.25$ °C warmer than today approximately _____
- How much warmer/colder was Greenland during the Minoan Warm Period (1300 BC) compared to today? $-(2.75-0.75) = 2$ °C warmer than today approximately _____
- Connect the theme of the various house-point questions so far with a point made in the first lecture – Think *independently* using data/numbers, to avoid brainwashing

1.

$$f_2(x, y, z)$$

$$:= (x+y)(y+z)(z+x)$$

$$:= (y+x)(y+z)(z+x)$$

$$= (y+xz)(x+z)$$

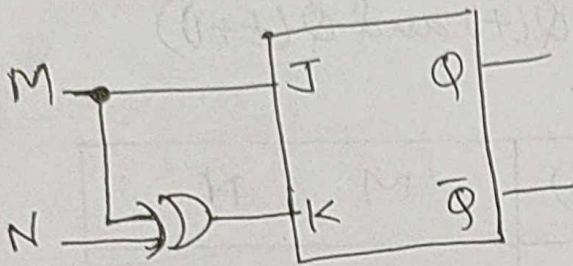
$$= xy + xz + xz + yz$$

$$= xy + yz + zx$$

$$= f(x, y, z)$$

Q.2]

→ MN flip-flop : $J = M$; $K = N \oplus M$.



State-transition table

M	N	J	K	$Q(t+1)$ [from JKFF]
0	0	0	0	$Q(t)$
0	1	0	1	0
1	0	1	1	$Q'(t)$
1	1	1	0	1

Excitation table

$Q(t)$	$Q(t+1)$	M	N
0	0	0	X
0	1	1	X
1	0	0 —OR— 1	1 0
1	1	0 —OR— 1	0 1

With the help of excitation table, we can model D flip-flop's state transition table in the form of M and N. (by observing $Q(t)$ and $Q(t+1)$)

D	$Q(t)$	$Q(t+1)$	M	N
0	0	0	0	X
0	1	0	$\frac{0}{1}$ OR $\frac{1}{0}$	
1	0	1	1	X
1	1	1	$\frac{0}{1}$ OR $\frac{0}{1}$	

Using K-map, we find 4 solutions for each pair of MN combinations

[1]

for M:

	$Q(t)$	0	1
D	0	0	0
1	1	1	0

$$M = D \cdot \overline{Q(t)}$$

D	$Q(t)$	M	N
0	1	0	1
1	1	0	0

for N:

	$Q(t)$	0	1
D	0	X	1
1	1	X	0

$$N = \overline{D}$$

[2]

for M:

	$Q(t)$	0	1
D	0	0	0
1	1	1	1

$$M = D$$

D	$Q(t)$	M	N
0	1	0	1
1	1	1	1

for N:

	$Q(t)$	0	1
D	0	X	1
1	1	X	1

$$N = 1$$

3

D	Q(t)	M	N
0	1	1	0
1	1	0	0

for M:

	D	Q(t)	
	0	1	
0	0	0	1
1	1	1	0

for N:

	D	Q(t)	
	0	1	
0	0	0	0
1	1	1	0

$$M = D \overline{Q(t)} + \overline{D} Q(t)$$

$$M = D \oplus Q(t)$$

$$N = D \overline{Q(t)}$$

4

D	Q(t)	M	N
0	1	1	0
1	1	1	1

for M:

	D	Q(t)	
	0	1	
0	0	0	1
1	1	1	1

for N:

	D	Q(t)	
	0	1	
0	0	0	0
1	1	1	1

$$M = D + Q(t)$$

$$N = D$$

Q.3]

→ The sequence is: $000 \rightarrow 010 \rightarrow 111 \rightarrow 100 \rightarrow 011$

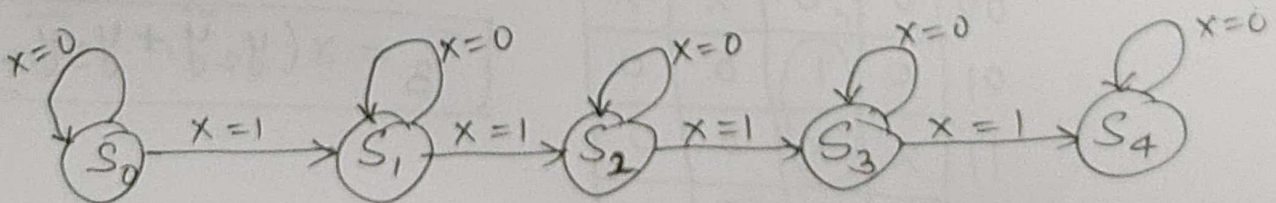
The state table is as follows:

S_0	000
S_1	010
S_2	111
S_3	100
S_4	011

The next-state map is as follows.

Lets say x is an input variable.

$Q(t)$	$Q(t+1)$	
	$x=0$	$x=1$
S_0	S_0	S_1
S_1	S_1	S_2
S_2	S_2	S_3
S_3	S_3	S_4
S_4	S_4	S_0



Now, the truth table of T flip flop is:

T	Q(t)	Q(t+1)
0	0	0
0	1	1
1	0	1
1	1	0

To derive the inputs of T flip-flops, we look at the next-state map and truth table of T flipflop.

Q(t) y ₃ y ₂ y ₁	Q(t+1)		T F/F inputs					
	x=0		x=0			x=1		
	Q ₃ Q ₂ Q ₁	Q ₃ Q ₂ Q ₁	T ₃	T ₂	T ₁	T ₃	T ₂	T ₁
0 0 0	0 0 0	0 1 0	0	0	0	0	1	0
0 1 0	0 1 0	1 1 1	0	0	0	1	0	1
1 1 1	1 1 1	1 0 0	0	0	0	0	1	1
1 0 0	1 0 0	0 1 1	0	0	0	1	1	1
0 1 1	0 1 1	0 0 0	0	0	0	0	1	1

Using K-map, we find:

for T₃:

y ₃ y ₂ \ y ₁ x	00	01	11	10
00	0	0	x	x
01	0	1	0	0
11	x	x	0	0
10	0	1	x	x

$$T_3 = y_2 \bar{y}_1 x + y_3 \bar{y}_2 x$$

$$T_3 = x(y_2 \bar{y}_1 + y_3 \bar{y}_2)$$

for T_2 :

$y_3 y_2 \backslash y_1 x$	00	01	11	10
00	0	1	X	X
01	0	0	1	0
11	X	X	1	0
10	0	1	X	X

$$T_2 = y_1 x + \bar{y}_2 x$$

for T_1 :

$y_3 y_2 \backslash y_1 x$	00	01	11	10
00	0	0	X	X
01	0	1	1	0
11	X	X	1	0
10	0	1	X	X

$$T_1 = y_2 x + y_3 x$$